A Clinically Guided Graph Convolutional Network for Assessment of Parkinsonian Pronation-Supination Movements of Hands

Zheng Xie, Rui Guo, Chencheng Zhang, and Xiaohua Qian

***Abstract*— In the clinically widely used rating scale (MDS-UPDRS), the pronation-supination movement task of hands is required for assessment of bradykinesia, which is a typical clinical symptom of Parkinson's disease (PD). Due to inter-rater variability in the task rating process, objective automated rating models are critically needed. Still, the performance of such models would be limited if prior knowledge of the clinical rating principles is not adequately accounted for. Therefore, we propose a clinically guided method which fully exploits the MDS-UPDRS rating principles to achieve consistent and accurate automated rating. First, a multi-scale framework which employs two graph convolutional networks (GCNs) as two streams is developed to extract transient and persistent features related to these rating principles. In particular, abnormal transient features are detected through a specialized multiple-instance-learning GCN. Moreover, the multiple-instance-learning GCN is equipped with an accumulation-aware ordered multiple-instance pooling module, which estimates sample-level severity by accounting for both the intra-instance intrinsic severity order and the inter-instance accumulation effects. Besides, an instance context encoding module is designed to combine the phase information in the pronation-supination movement cycles with instances' motion features. This facilitates the differentiation between PD-induced halts and natural periodic halts. Our method demonstrated excellent performance on both a large clinical dataset and an additional independent test dataset. Our proposed scheme only requires consumer-level cameras, and therefore exhibits high potential for large-scale applications in PD telemedicine.**

***Index Terms*— Graph convolutional network, multiple instance learning, Parkinson's disease, pronation-supination movements of hands, video-based assessment.**

I. Introduction

PARKINSON'S disease (PD) is a common neurodegenerative disease, which brings inconvenience and even risks to patients' daily lives [1]. At present, the Movement Disorder Society-Sponsored Revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS) is the most widely used clinical rating scale for PD assessment [2], [3]. Based on this rating scale, the severity of PD motor symptoms is assessed through the patient's performance on various motor function tasks. In particular, the pronation-supination movement task of hands (the PS task, for short) is vital for the assessment of bradykinesia, a typical PD symptom whose severity and extent are often used to judge the effects of dopaminergic medications and deep brain stimulation [4].

In the PS task, the patient is required to turn his/her palm up (the supination orientation) and down (the pronation orientation) repeatedly as fast and as fully as possible. A score between 0 and 4 is given according to the variations in the movement speed, amplitude, hesitations, and halts. Due to the subjective nature of human ratings, the scores tend to vary significantly among different neurologists [3]. Therefore, automated rating systems are strongly needed in clinical practice to obtain consistent scores.

Efforts have been made towards automated PS-task assessment [5], [6], [7]. For example, using three-dimensional (3D) LeapMotion sensor data taken during the PS movements, Moshkova et al. [5] employed a convolutional neural network (CNN) as a feature extractor and a support vector machine (SVM) as a classifier to recognize PD patients. Dror et al. [6] introduced a method for palm detection in raw data of 3D depth sensors, and then trained a PD detector based on features of speed, acceleration, amplitude, and frequency from the PS movements. However, most of these studies chose just returning binary decisions (whether symptom is present or not) instead of giving a subtler five-point classification of symptom severity as required by the MDS-UPDRS [5], [6]. Moreover, since these systems are based on contact sensors [7] or 3D

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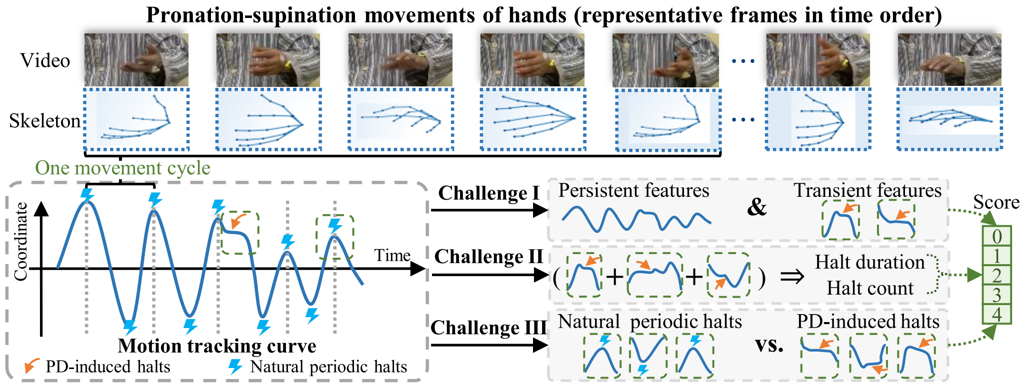

Fig. 1. Three challenges to be addressed in our assessment framework for the pronation-supination movement task of hands: I) Simultaneous extraction of both persistent and transient features; II) Within a multiple-instance-learning scheme, instances (i.e., clips) are naturally ordered by their intrinsic severity levels, and the sample-level severity increases as more clips include PD-induced halts; III) Instances lack features of background periodic pronation-supination (PS) movements, and this leads to confusion between periodic halts and PD-induced halts.

optical motion-capture systems [5], [6], their practicality is limited. In fact, the contact sensors may interfere with the patient movements during motion tasks, and thereby lead to inaccurate assessment results. On the other hand, the optical motion-capture systems are expensive and of inadequate precision [8].

By contrast, as the video-based systems are contactless and have low device requirements, these systems are generally more suitable for large-scale clinical applications. At present, such video-based systems have demonstrated remarkable performance in motor function assessment for various MDS-UPDRS tasks other than the PS task, such as gait [9], toe tapping [10], arising from chair [11], and finger tapping [12]. Therefore, we decide to achieve automated five-point rating of the PS task through a video-based approach in this study.

Human skeleton data (i.e., sets of human joint coordinates) can be extracted from red-green-blue (RGB) videos through pose estimation algorithms. In comparison to raw RGB video data, this skeleton data is more robust against background changes [13], and therefore has been widely used in human action recognition tasks. Among the existing skeleton-based action recognition models, those based on graph convolutional networks (GCNs) can effectively exploit the intrinsic topology of the human skeleton data in order to boost the recognition performance. Indeed, GCNs have already demonstrated superior outcomes in various PD motion assessment tasks [14], [15], [16]. Consequently, we decide to build our method upon skeleton data extracted from videos and adopt GCN as a key building block of our automated rating model.

However, there are still several challenges that need to be handled by the proposed automated rating system (Fig. 1). First of all, according to the MDS-UPDRS, both persistent and transient motion features need to be captured for the PS task (Fig. 1, challenge I). Specifically, the MDS-UPDRS ratings are based on slow-changing characteristics which can be typically observed over the whole process (such as drops in speed or amplitude), as well as sudden and sporadic characteristics (such as hesitations and halts). The transient motion abnormalities are vital for severity staging and are distributed discretely across time. However, by applying global pooling operations after temporal convolution, most existing action recognition models can be effective in capturing persistent features but not the transient ones [17].

To simultaneously extract transient and persistent features, we design a multi-scale framework with two streams, namely the transient stream and the persistent stream. To effectively capture transient features, we introduce a multiple-instance-learning (MIL) scheme into the transient stream and design a MIL-based GCN. By dividing each whole sequence of skeleton data (i.e., a bag) into temporally short clips (i.e., instances), the model can focus better on small-scale transient features. Features are extracted at the clip level, and are finally aggregated to yield sample-level predictions. We also adopt a traditional global-pooling-based GCN in the persistent stream to extract persistent features from whole skeleton sequences.

Our multi-scale framework generally leads to accurate PS-task assessment through the combination of the two streams. However, under the MIL-based framework of the transient stream, we encounter another two key challenges, as shown in Fig. 1 (challenge II and III).

One challenge is that each MIL clip has its own intrinsic severity, and cumulatively contributes to the sample-level severity (Fig. 1, challenge II). This poses challenges for multiple-instance aggregation modeling. Specifically, 1) clips can be ordered by their intrinsic severity (i.e., the durations of the PD-induced halts), and the sample-level severity is mainly determined by the high-severity clips. 2) The sample-level severity increases with the accumulation of clips containing PD-induced halts. Existing multiple-instance aggregation methods (such as max-pooling, average-pooling or weighted summation) fail to model the order of the intrinsic severities of different clips and ignore the clip accumulation effects on the sample-level severity. This leads to suboptimal multiple-instance aggregation performance.

Another challenge is that MIL clips lack information about background periodic movements (Fig. 1, challenge III). This shortcoming leads to confusion between natural periodic halts and PD-induced halts. In fact, periodic halts that occur at the limiting positions of pronation or supination movements have motion features similar to those of PD-induced halts. Without knowledge of the corresponding phase of background periodic movements, the PD-induced halts are usually confused with natural periodic ones, and this leads to difficulties in transient feature extraction.

To meet the MDS-UPDRS requirements in multiple-instance aggregation, the intra-instance intrinsic severity order is explicitly modeled to obtain key high-severity clips. Then, key clips are aggregated through an accumulation-aware process, where higher numbers of abnormal clips possibly lead to a higher sample-level severity. Besides, the phase information of the periodic movements is extracted and embedded into the MIL clips to enhance the discriminability between natural periodic halts and PD-induced halts.

In general, we propose a clinically guided framework with a multiple-instance GCN for automated PS-task assessment, where MDS-UPDRS prior knowledge is accounted for in order to effectively model how neurologists rate the PS task. Specifically,

1) Hand skeleton sequences are extracted from videos through a human pose estimation algorithm, and are then fed into a GCN-based model to obtain scores for the PS task.

2) A multi-scale framework with two streams is proposed, where a global-pooling-based GCN suitable for large-scale persistent features extraction is combined with another MIL-based GCN dedicated to extracting small-scale transient features. Thus, the proposed framework can effectively capture both persistent and transient features.

3) An accumulation-aware ordered multiple-instance pooling module is designed in order to explicitly assign severity orders to clips and enable the estimation of sample-level severity based on accumulated clips. Thereby, the multiple-instance aggregation process can be modeled according to the MDS-UPDRS rating principles.

4) An instance context encoding module is proposed to extract phase information of the background periodic movements and then embed this information into MIL clips. This helps the model to discriminate between natural periodic halts and PD-induced halts.

Our key contributions can be summarized as follows. Practically, a video-based automated rating system is proposed for objective assessment of the pronation-supination movements of hands, which is highly practical with low device requirements (only a cheap consumer-level camera). Technically, the proposed model is designed to fully exploit prior clinical knowledge of the MDS-UPDRS rating principles, and can serve as a framework for bradykinesia assessment in PD patients. Specifically,

1) A multi-scale framework with an integrated multiple-instance network is proposed to effectively capture both persistent and transient motion features.

2) An accumulation-aware ordered multiple-instance pooling module is developed to model the intrinsic order of clip instances and the cumulative process of bag-level labeling.

3) An instance context encoding module is proposed to embed phase information of periodic movements into clip instances. This facilitates the differentiation between non-periodic PD-induced halts and normal periodic halts.

II. Related Work

A. Human Pose Estimation in Motion Tracking

Human pose estimation can be broadly categorized into top-down and bottom-up approaches. In top-down approaches [18], [19], [20], bounding boxes are first predicted for human bodies, and then single-person pose estimation is applied within these bounding boxes to obtain joint coordinates. For example, Fang et al. [19] devised a spatial transformer to mitigate the reverse effect of imperfect bounding boxes on single-person pose estimation. Papandreou et al. [20] proposed a method to simultaneously predict disk-shaped heatmaps and offset fields, thereby enhancing the precision of joint localization. On the other hand, in bottom-up approaches [21], [22], [23], joint positions are initially predicted, and these joints are subsequently assembled to form individual human poses. For example, Pishchulin et al. [21] designed integral linear programming for joint-to-person association. Cao et al. [22] proposed part affinity fields to encode the directions of limbs, which enables faster and more accurate joint-to-person association.

In our study, human pose estimation is adopted as an essential preprocessing step to extract skeletons from videos, and available state-of-the-art well-trained models for human pose estimation are employed to ensure accurate results.

B. Graph Convolutional Networks (GCNs) in Skeleton-Based Action Recognition

In graph convolutions, recursive neighborhood aggregation is performed over node/edge representations. This process helps capture the graph topology and generates useful node representations. Various aggregation methods have been proposed to achieve effective graph convolution [24], [25], [26]. For example, Niepert et al. [24] proposed a method where nodes are assembled into a fix-sized neighborhood, and this neighborhood is then mapped into a fix-sized vector for processing by CNNs. Kipf et al. [25] replaced spectral graph convolution with a first-order approximation and further simplified the graph convolution into matrix multiplication.

GCNs have been extensively employed in skeleton-based human action recognition to leverage the latent information embedded in the human physical topology, ultimately boosting classification performance [17], [27], [28], [29], [30], [31], [32], [33]. To extract motion features in a unified way across time and space, Yan et al. [17] proposed a spatial-temporal GCN (ST-GCN) that combines graph convolution over the space dimension with convolution over the time dimension. Based on the ST-GCN, advanced graph convolutions have been proposed to enhance the GCN performance. Shi et al. [30] proposed a data-dependent adaptive graph structure to increase the GCN flexibility. Also, Shi et al. [31] modeled the skeleton structures as directed acyclic graphs in which the graph vertices and edges represent joints and bones, respectively. Given the impressive performance of GCNs in action recognition, we have chosen to adopt GCNs in our study to facilitate effective feature extraction on human skeletons.

C. Spatial-Temporal Feature Enhancement

With deep feature extraction methods becoming prevalent in computer vision [34], [35], spatial-temporal feature enhancement has been widely explored to improve feature representations in various video-based tasks, such as object detection [36], instance segmentation [37] and captioning [38], [39]. Spatial modeling often involves the use of feature pyramids to achieve multi-scale feature extraction [40], [41], [42], and also involves GCNs to capture spatial relationships [43]. For temporal modeling, temporal attention [38] and dilated convolution [39] have been incorporated in CNNs to extract long-term temporal relationships. Transformer-based models are also designed to naturally capture long-term relationships through self-attention [44]. Furthermore, cross-attention [45] has been applied to enhance pairwise relationship modeling between temporal frames.

In skeleton-based action recognition, the physical topology of the human skeleton is effectively leveraged through spatial graph convolutions in GCNs. To further improve recognition performance, novel temporal operations have been proposed to replace CNNs for enhancing temporal feature extraction. For example, Liu et al. [32] employed larger time windows through sampling, enabling the construction of long-range connections over the temporal dimension. Chen et al. [33] designed a multi-scale temporal convolution to extract temporal information on different scales. However, these studies mainly focus on extracting large-scale global information through global pooling schemes, neglecting small-scale transient features. Consequently, we develop a GCN under the multiple-instance-learning scheme as an additional stream in our framework to capture these transient features.

D. Multiple-Instance Classification

In standard multiple-instance classification problems, a dataset $\mathcal{X} = \{X_i, 1 \leq i \leq M\}$ is composed of M bags, where each bag $X_i = \{x_{ij}, 1 \leq j \leq N_i\}$ contains N_i instances. Each instance x_{ij} has its own binary label $y_{ij} \in \{0, 1\}$ which is not given in the dataset, and only the bag-level label $Y_i \in \{0, 1\}$ is known. The relationship between the labels of a bag and its instances can be expressed as:

$$Y_i = 1 - \prod_j (1 - y_{ij}) \tag{1}$$

In deep learning approaches, such a relationship is modeled through multiple-instance pooling, where instance-level results are aggregated to obtain bag-level predictions. Existing multiple-instance pooling methods can be classified into embedded-space and instance-space methods. The embedded-space methods perform instance aggregation in the feature space, and include methods of feature concatenation [46] and attention-based summation [47]. The instance-space methods aggregate instances based on their classification probabilities or labels, and include methods of max-pooling [48], noisy-OR networks [49], and adaptive pooling [50].

In medical image analysis, multiple-instance frameworks are widely used to generate image-level prediction from patches [46], [49], [51]. Improved performance and interpretability can be achieved by employing suitable multiple-instance pooling methods. Considering that different patches may have different levels of influence on the final image-level predictions, Wang et al. [46] applied attention-based multiple-instance pooling in gastric cancer diagnosis, where key patches can be identified with large attention weights. To diagnose pulmonary nodule cancer, Liao et al. [49] proposed a leaky noisy-OR multiple-instance pooling layer in order to model the uncertainty in determining the overall cancer probability based on the malignancy level of multiple pulmonary nodules. However, these methods do not quite fit the aggregation process with the ordered and cumulative characteristics described in the MDS-UPDRS guidelines. Therefore, a novel multiple-instance pooling method is proposed to aggregate clips according to the MDS-UPDRS clinical rating principles in order to boost the performance on the PS-task assessment.

III. Methodology

As shown in Fig. 2, the proposed model is composed of the GCN-based persistent and transient streams (Section III-B), where persistent and transient features are extracted, respectively. The two streams produce classification logits (classification probabilities before softmax), which are fused to obtain the overall predictions. In the transient stream, transient features are extracted from clips under a multiple-instance-learning framework, where an instance context encoding module (Section III-D) is employed to embed phase information of the background periodic movements into the clips. After feature extraction at the clip level, an accumulation-aware ordered multiple-instance pooling module (Section III-C) is applied to aggregate clip features and generate classification logits in the transient stream.

A. Background Movement Extraction

Pronation-supination movement task of hands can be viewed as a combination of foreground movements of transient PD-induced halts and background periodic movements. The periodic movement signals are essential for the extraction of both large-scale persistent features in the persistent stream (Section III-B) and phase information in the instance context encoding module (Section III-D). However, due to the RGB frame blur caused by fast motion, some joint data points are missing in the skeleton data obtained from RGB frames, and this causes difficulties in extracting background periodic movements. To handle these difficulties and process signals with missing data points, Bayesian temporal factorization, a time-series method based on low-rank matrix decomposition and vector autoregressive modeling [52], is employed to obtain stable and smoothed signals representing periodic movements from skeleton data.

B. Multi-Scale Framework

To simultaneously extract persistent and transient features, a multi-scale framework is designed with two streams (the persistent stream and the transient stream), where spatial-temporal graph convolution layers (as proposed in ST-GCN [17]) are adopted for feature extraction. Each spatial-temporal graph

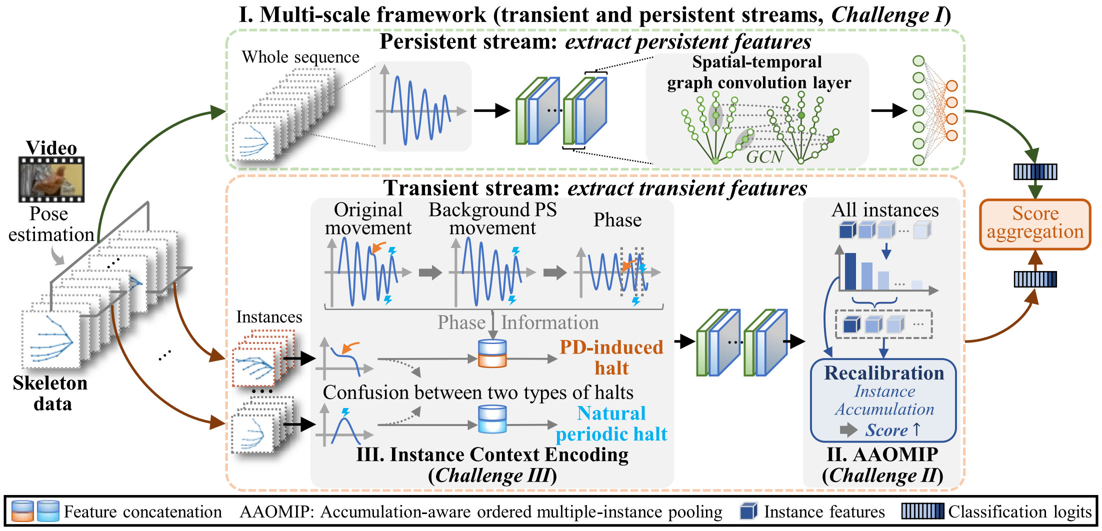

Fig. 2. The overall architecture of the proposed clinically guided framework with a multiple-instance GCN. The framework has two key network components: the whole-sequence-based persistent stream and the clip-based transient stream. For the transient stream, clip features are combined with phase information of the background periodic pronation-supination (PS) movements in the instance context encoding (ICE) module, and are finally aggregated through an accumulation-aware ordered multiple-instance pooling (AAOMIP) module for sample-level severity estimation.

convolution layer consists of a graph convolution over spatial joints and a convolution over temporal frames, enabling the extraction of spatial-temporal features. In our work, a skeleton sequence can be represented as $X \in \mathbb{R}^{C \times T \times V}$, where C, T, and V represent the number of channels (initially the number of coordinates), frames, and joints, respectively. The graph convolution on skeleton $X_t \in \mathbb{R}^{C \times V}$ for frame t can be described as:

$$gc\left(X_t\right) = \sum\nolimits_{s=1}^{S} W X_t (\tilde{A}_s \odot M_s) \tag{2}$$

where $W \in \mathbb{R}^{C' \times C}$ is a learnable feature transformation, $\tilde{A}_s$ is the normalized adjacency matrix, and M_s is the learnable edge importance [17]. Additionally, $\odot$ represents an element-wise product, and S is set to 2, as distance partitioning is used [17]. The design of the afore-mentioned two streams using spatial-temporal graph convolution layers is described as follows.

1) The Persistent Stream: This stream focuses on large-scale persistent features. It takes background movement signals extracted by the Bayesian temporal factorization method as input. The input signals are then processed by 7 spatial-temporal graph convolution layers. After that, the obtained feature representations are global average pooled and fed into a fully connected layer to obtain classification logits that represent symptom severity in terms of persistent features.

2) The Transient Stream: This multiple-instance-learning-based stream focuses on small-scale transient features. Multiple overlapped clips (i.e., instances) are generated by slicing each whole skeleton sequence (i.e., a bag) through a sliding window. For each sequence X, its clips $\left\{x_j \in R^{C \times L \times V}\right\}$ are taken as the input of the transient stream, where L is the number of frames for each clip. After preliminary motion feature extraction through a spatial-temporal graph convolution layer, the background movement information is aggregated into clip features through the instance context encoding module (Section III-D). Afterwards, the clip features

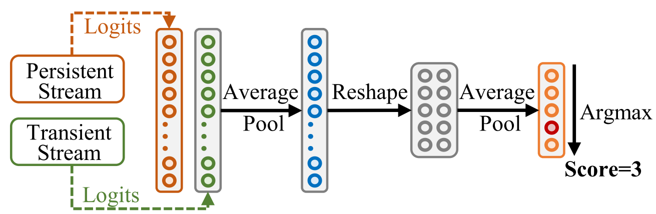

Fig. 3. The process of score aggregation, which generates final classification logits from the persistent stream and the transient stream.

are passed through 6 spatial-temporal graph convolution layers and finally aggregated by an accumulation-aware ordered multiple-instance pooling module (Section III-C) to generate sample-level classification logits, which represent the symptom severity in terms of transient features.

Classification logits from both streams are then aggregated into a sample-level score. The aggregation process is shown in Fig. 3. Average pooling is adopted for combining the output logits of the two networks and also for adapting the level number of the combined logits as MDS-UPDRS severity levels. Specifically, the output logits of the transient stream and the persistent stream (denoted respectively as $p_{ID} = [p_{ID}^1, \ldots, p_{ID}^K]$ and $p_{BG} = [p_{BG}^1, \ldots, p_{BG}^K] \in \mathbb{R}^K$) are first average-pooled to yield sample-level classification logits $p_{AGG} = \left[p_{AGG}^1, \ldots, p_{AGG}^K\right]$ according to (3), where K, the level number of p_{ID} and p_{BG}, is set to be larger than the number of severity levels K_0 in order to retain more information. Afterwards, the logits in p_{AGG} are divided into K_0 groups and averaged within each group according to (4), which yields the final prediction logits $p = \left[p^1, \ldots, p^{K_0}\right]$.

$$p_{AGG}^k = \frac{1}{2}\left(p_{ID}^k + p_{BG}^k\right), \quad 1 \leq k \leq K \tag{3}$$

$$p^k = \frac{1}{G}\sum\nolimits_{i=(k-1)\cdot G+1}^{k\cdot G} p_{AGG}^i, \quad 1 \leq k \leq K_0;$$
$$G = \left\lfloor \frac{K}{K_0} \right\rfloor \tag{4}$$

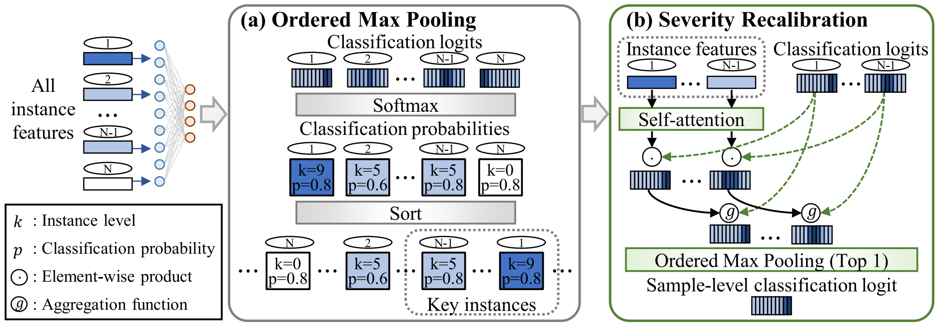

Fig. 4. The architecture of the accumulation-aware ordered multiple-instance pooling (AAOMIP) module, which is composed of two steps: (a) ordered max pooling and (b) severity recalibration.

C. Accumulation-Aware Ordered Multiple-Instance Pooling (AAOMIP)

We propose an accumulation-aware ordered multiple-instance pooling (AAOMIP) module in order to model the multiple-instance aggregation process based on the MDS-UPDRS rating principles. According to the MDS-UPDRS, the sample-level severity is related to both the halt duration and count. For example, samples containing 1-2 halts should be rated with a score of 1, while samples containing 3-5 halts should be rated with a score of 2, and samples which either contain a longer (i.e., more severe) halt or more than 5 halts should be rated with a score of 3 [3]. Therefore, the proposed pooling method should have the following characteristics:

Characteristic 1: Instances are ordered by their intrinsic severity, and the bag-level severity is mainly determined by high-severity instances.

Characteristic 2: The bag-level severity is related to the instance-level severity distribution, and accumulation of instances may lead to an increase in the bag-level severity.

The AAOMIP module is proposed to explicitly model the above two characteristics. Characteristic 1 is modeled through ordered max pooling, which explicitly considers the severity-based instance order and identifies key high-severity instances. To model characteristic 2, a severity recalibration mechanism is designed to simulate the relationship between the accumulation of instances and the increase in the bag-level severity. The AAOMIP architecture is depicted in Fig. 4.

All instance features are first fed into an instance-level classifier to determine the severity of each instance. Specifically, the feature vector v_j of the instance x_j is fed into the instance-level classifier, which yields output logits $o_j = [o_j^1, \ldots, o_j^K] \in \mathbb{R}^K$, $1 \le j \le N$, where N is the number of instances in sample X. Afterwards, two operations for AAOMIP are performed, as follows:

Step 1: The ordered max pooling is used to sort instances based on their severity. The instances $\{x_j, 1 \le j \le N\}$ of the sample X are first sorted in a descending order according to their predicted severity level $l_j = argmax_{1 \le k \le K} p_j^k$, where $p_j = [p_j^1, \ldots, p_j^K] = softmax(o_j)$ is the normalized classification probability. We denote by X^{k_0} the set of instances of the sample X which are classified into level k_0. Such instances are further sorted in a descending order according to their respective $p_j^{k_0}$ values. Afterwards, the instances at any two positions $p, q (p < q)$ should satisfy either (5a) or (5b).

$$l_p > l_q \tag{5a}$$
$$l_p = l_q = k_0; \quad p_p^{k_0} > p_q^{k_0} \tag{5b}$$

The top instances have higher predicted severity levels or higher probabilities to be labelled with these severity levels, and are considered to have a larger influence on the bag-level severity. Conversely, most low-ranking instances with lower severity levels are expected to contain no transient movement abnormality, and thus have small or no influence on the bag-level severity. The top N^{key} instances are then taken as the set of key instances X^{key}, and this set is later used to determine the bag-level severity. Here, $N^{key} = P \cdot N$, where P controls the proportion of key instances.

Step 2: The severity recalibration is employed to aggregate instances within the set of key instances X^{key} in order to obtain bag-level predictions. During the aggregation process, the accumulation of instances monotonically leads to an increase in the bag-level severity. To model the accumulation of instances, a self-attention mechanism [53] is used to enable communications among instances. Specifically, the feature vectors $\{v_j, 1 \le j \le N^{key}\}$ of all instances within X^{key} are fed into a self-attention layer in order to obtain an alternative set of feature vectors $\{v'_j, 1 \le j \le N^{key}\}$, whose members v'_j incorporate information from other instances. Afterwards, v'_j is passed through a recalibration factor generation network f to obtain the recalibration factor $r_j = f\left(v'_j\right)$. Then, r_j is used to produce $p'_j = r_j \odot p_j$. Consequently, p'_j embeds the influence of other instances, and thus has the potential to be viewed as the result of instance accumulation. To further model the monotonicity between instance accumulation and the increase in the bag-level severity, we apply an aggregation function g to p_j and p'_j to get the recalibrated classification logits p''_j. The function g is defined as in (6) to ensure that p''_j has a higher severity level than those of p_j and p'_j.

$$p''^k_j = g\left(p_j, p'_j\right)^k = \begin{cases} \max\left(p_j^k, p'^k_j\right), & k \ge l_j \\ \min\left(p_j^k, p'^k_j\right), & k < l_j \end{cases} \tag{6}$$

Finally, the classification logit outputs p_{ID} for the transient stream are produced by applying ordered max pooling with

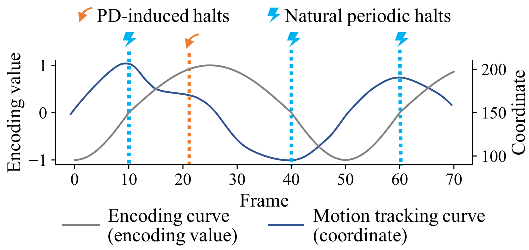

Fig. 5. Instance context encoding. A whole movement cycle (Frames 11 to 60) includes a period of hand pronation (Frames 11 to 40) and a period of hand supination (Frames 41 to 60). The two periods can be respectively represented by the positive and negative halves of a sine signal in the generated phase encoding.

$P = 1/N^{key}$ to the set of the recalibrated classification logits $\{p''_j, 1 \leq j \leq N^{key}\}$ (i.e., choosing the top 1 item).

D. Instance Context Encoding (ICE)

Natural periodic halts possess similar movement patterns with PD-induced halts and therefore may interfere with the detection of PD-induced halts in the transient stream. We designed an instance context encoding (ICE) module which fuses the phase of the background periodic movements into clips to facilitate discrimination between two types of halts.

Since natural periodic halts only occur at the limiting positions of the pronation-supination movements of hands, while PD-induced halts happen between these limiting positions, the phase of the pronation-supination cycles can be used to determine whether a halt is induced by PD. Therefore, phase encoding indicating the cycle phase is generated from the background movement signals (Section III-A), and added to the clip features. Specifically:

The mean values of the x coordinates of four joints on the index finger is calculated to represent the background periodic movements (plotted as the motion tracking curve in Fig. 5). The local minima and maxima of such mean values correspond to the limiting positions of the pronation-supination movements of hands, and a movement cycle is defined as the interval between two neighboring local maxima. Frame indexes for local maxima $\{t_n^{max}\}$ and minima $\{t_n^{min}\}$ are then obtained by applying automatic multiscale-based peak detection [54] to the mean-value signals. Interpolation is performed if necessary to ensure the alternative appearance of the local maxima and minima. Based on the extracted extreme points (including local maxima and minima), sine signals with phase ranges of $[0, \pi]$ and $[\pi, 2\pi]$ are used to indicate the movement from a local maximum to a local minimum and that from a local minimum to a local maximum, respectively. If the first extreme point of the mean values is assumed to be a local maximum, then the phase encoding $E \in \mathbb{R}^T$ at frame t could be computed as:

$$E_t = \begin{cases} \sin\left(\dfrac{\pi(t - t_n^{max})}{t_n^{min} - t_n^{max}}\right), & t_n^{max} \leq t < t_n^{min} \\ \sin\left(\dfrac{\pi(t - t_n^{min})}{t_{n+1}^{max} - t_n^{min}} + \pi\right), & t_n^{min} \leq t < t_{n+1}^{max} \end{cases} \tag{7}$$

TABLE I
CLINICAL PATIENT INFORMATION IN THE CROSS-VALIDATION DATASET

Characteristics	Value
Number of Videos	399
MDS-UPDRS score distribution (score-0/1/2/3/4)	109/128/76/62/24
Sex (F/M)	40/87
Age (mean±std)	63.62±10.79

TABLE II
CLINICAL PATIENT INFORMATION IN THE INDEPENDENT TEST DATASET

Characteristics	Value
Number of Videos	91
MDS-UPDRS score distribution (score-0/1/2/3/4)	28/23/20/14/6
Sex (F/M)	12/36
Age (mean±std)	61.64±10.69

Phase encoding can be computed analogously when the first extreme point is a local minimum. Finally, the phase encoding (Fig. 5) is concatenated to the clip features for subsequent feature processing. According to (7), natural periodic halts always correspond to the phase 0 or π of the encoding signal, where the encoding value is 0 (i.e., at blue vertical lines in Fig. 5), while PD-induced halts correspond to phases other than 0 or π (e.g., at the orange vertical line in Fig. 5). By introducing phase encoding into clip features, these two types of halts can be easily discriminated by the transient stream.

IV. Datasets and Experiments

A. Datasets

The datasets used in this study were provided by the Neurosurgery Department of Ruijin Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China. These datasets include a cross-validation dataset (Table I) and an independent test dataset (Table II). Both datasets contain videos of PD patients performing the PS task under different conditions (with/without medication, before/after surgery). Therefore, several videos are typically available for each patient. Videos in the cross-validation dataset were collected from 2017 to 2019, while those in the independent test dataset were collected in 2020. The videos were recorded with a frame size of 1080 × 1920 and a frame rate of 30 frames per second. The task performance in each video was assessed with a score between 0 and 4 according to the MDS-UPDRS guidelines.

To acquire precise hand skeletons, we employed a two-step skeleton extraction process. Preliminary pose detection in the collected videos was firstly performed to extract whole-body skeletons using ‘OpenPose’ (v1.7.0) [22], [55], a bottom-up pose detection library known for its increased robustness against background interference. From the results of pose estimation, the range of the pronation-supination movements of hands was calculated. After that, based on this range, hand motion videos containing only hand movements were spatially cropped from the original videos. Finally, hand skeletons were estimated in the cropped hand motion videos using a well-trained specialized hand skeleton extraction model provided by the ‘MMPose’ [56] toolbox. This model is based on the HRNetv2 [57] and DarkPose [58] frameworks and has been well trained on the CMU Panoptic HandDB dataset [55]. We horizontally flipped all samples where the task is performed with the left hand to standardize the input features.

B. Implementation Details

A 5-fold cross-validation (CV) scheme is adopted in our experiments, where all videos from the same patient are assigned to a single fold in order to avoid data leakage and better demonstrate performance stability when unseen patients are encountered. The reported evaluation metrics include the accuracy (Acc), recall (Rec), precision (Pre), F1-score (F1), and the area under the receiver operating characteristic (ROC) curve (AUC). Since a 1-point score difference is clinically acceptable in MDS-UPDRS-based PD assessment [59], we further include the acceptable accuracy (Acceptable Acc) as an additional metric as has been done in some previous studies [10], [11], [14], [16]. The acceptable accuracy is calculated as:

$$Acceptable\ Acc = N_{|y-y_0|\leq 1}/N \tag{8}$$

where $N_{|y-y_0|\leq 1}$ denotes the number of samples whose predicted scores y are no more than 1 point away from the ground-truth scores y_0, and N denotes the total number of samples. We report results averaged over three repetitions of 5-fold CV.

The frame count of the background movement signals is normalized to 300 by clipping or zero padding before feeding to the persistent stream. For the transient stream, clips from the original skeleton sequences are used as inputs.

A simple oversampling scheme is designed to tackle the class imbalance problem. Specifically, samples of different classes are divided into class-specific pools. For each training epoch, a fixed number of samples (100 in our experiments) is randomly selected from each pool without replacement. Once a pool becomes empty, it is refilled with all samples of the corresponding class. All experiments were conducted on the PyTorch deep learning framework with a GTX1080Ti GPU. Training was carried out with the cross-entropy loss, the stochastic gradient descent (SGD) optimizer with a momentum of 0.9, a total of 80 training epochs, and an initial learning rate of 0.01 (decreased to 0.001 at the 64^{th} epoch).

In our study, the levels of the classification logits given by the two streams were empirically set to $K = 10$. Also, a clip length of 16 frames ($L = 16$) and a stride of 8 frames were used for clip generation for the transient stream. The proportion of the key clips in the accumulation-aware ordered multi-instance pooling module was set to 50% ($P = 0.5$).

C. Classification Results

We evaluated the performance of our proposed model on the cross-validation dataset. Figure 6 shows the associated confusion matrix and ROC curves for the classification results of each score. Our model has an overall accuracy of 61.11% and an overall acceptable accuracy of 91.85%. Also, our model attained an accuracy over 40% and an acceptable accuracy over 90% (Table III) for all scores except score 4. The lower performance on the clinically rare score of 4 is mainly due to the insufficiency of the training samples.

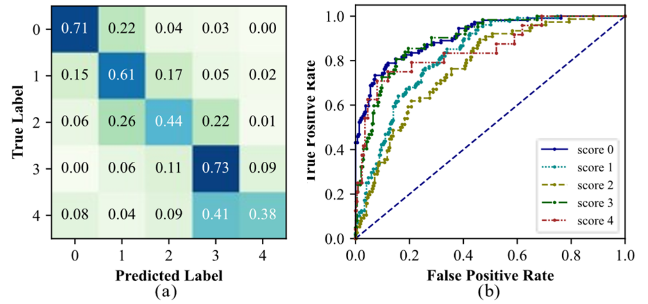

Fig. 6. Scoring results of our proposed model on the cross-validation dataset: (a) the confusion matrix, and (b) the ROC curves.

TABLE III
SCORING RESULTS ON THE CROSS-VALIDATION DATASET

Scores	Acc (%)	Acceptable Acc (%)	Rec (%)	Pre (%)	F1 (%)	AUC
Score 0	70.65	92.74	70.65	75.23	72.37	0.91
Score 1	60.63	92.61	60.63	62.02	61.02	0.82
Score 2	44.46	92.90	44.46	49.47	46.24	0.79
Score 3	72.82	93.29	72.82	56.25	63.19	0.90
Score 4	38.07	78.94	38.07	46.79	38.73	0.86
ALL	61.11	91.85	57.33	57.95	56.31	0.86

TABLE IV
SCORING RESULTS FOR THE ABLATION EXPERIMENTS WITH THE PROPOSED MODULES (PSS: PERSISTENT STREAM, TSS: TRANSIENT STREAM, AAOMIP: ACCUMULATION-AWARE ORDERED MULTI-INSTANCE POOLING, ICE: INSTANCE CONTEXT ENCODING)

Models	Acc (%)	Macro average		
		Rec (%)	Pre (%)	F1 (%)
PSS (Baseline)	51.54	51.00	52.47	50.25
TSS	49.14	43.31	45.64	42.81
PSS+TSS	54.24	51.65	53.12	51.08
PSS+TSS+AAOMIP	59.18	57.15	57.53	55.97
PSS+TSS+ICE	55.24	52.19	54.12	51.45
Our model (PSS+TSS+AAOMIP+ICE)	**61.11**	**57.33**	**57.95**	**56.31**

D. Ablation Study

We conducted comprehensive ablation studies to demonstrate the effectiveness of each module in our proposed model. The ST-GCN architecture (with 5 layers and a distance strategy) [17] is adopted as our baseline model, and is also used as the persistent stream in our proposed model.

1) Multi-Scale Framework: To verify the effectiveness of the proposed multi-scale framework, we reported results before and after adding the transient stream to the baseline model (the persistent stream). Table IV shows that the multi-scale framework leads to accuracy improvements of 2.70% and 5.10% compared to the persistent stream and the transient stream, respectively. Through the multi-scale framework, the transient and persistent features can complement each other, leading to a boost in classification performance.

2) Accumulation-Aware Ordered Multi-Instance Pooling (AAOMIP): To demonstrate the effectiveness of the proposed AAOMIP module, we added this module to the multi-scale framework. This combination brought an increase in accuracy of 4.94% (Table IV). We also replaced the AAOMIP module in our proposed model with other multiple-instance

TABLE V
PERFORMANCE COMPARISON BETWEEN DIFFERENT MULTIPLE-INSTANCE POOLING METHODS

Paradigms	Pooling Methods	Acc (%)	Macro average		
			Rec (%)	Pre (%)	F1 (%)
Instance Space	Max Pooling [48]	55.24	52.19	54.12	51.45
	AAOMIP	**61.11**	**57.33**	**57.95**	**56.31**
Embedded Space	Multiple-Instance Attention [60]	58.13	55.92	54.68	54.02
	Average Pooling [48]	58.91	56.19	57.08	55.02

TABLE VI
SCORING RESULTS WITH DIFFERENT NUMBERS OF OUTPUT SEVERITY LEVELS FOR THE PERSISTENT AND TRANSIENT STREAMS

Severity Level	Acc (%)	Acceptable Acc (%)	Macro average		
			Rec (%)	Pre (%)	F1 (%)
5	57.98	90.78	55.17	54.44	53.16
10	**61.11**	91.85	57.33	**57.95**	56.31
15	59.41	**91.92**	55.46	55.84	53.96
20	58.76	91.51	**57.86**	57.34	**56.41**
30	58.33	91.14	55.79	56.21	54.62
40	58.62	91.39	56.31	56.46	55.05

pooling methods to investigate the relative strength of our module. Table V shows that the accuracy of the model equipped with AAOMIP is 5.87% higher than its counterpart using max pooling. Empirically, the embedded-space methods (such as multiple-instance attention and embedded-space average pooling) perform better than the instance-space pooling methods (such as max pooling) [60]. However, as an instance-space pooling method, the AAOMIP module surpasses multiple-instance attention and average pooling on all reported metrics (Table V). This further demonstrates the effectiveness of the proposed AAOMIP module.

3) Instance Context Encoding (ICE): To experimentally show the effectiveness of the proposed ICE module in reducing the interference from natural periodic halts, we added the ICE module to the multi-scale framework with/without the AAOMIP module. In both cases, an increase in all reported metrics can be observed (Table IV), and this confirms the significance and effectiveness of the proposed ICE module.

E. Parameter Sensitivity Analysis

1) Number of the Severity Levels for the Persistent and Transient Streams: Our proposed model generates final classification results based on the output severity levels of both the persistent and transient streams. We varied the number of severity levels to analyze its effect on the model performance. Table VI shows that when the number of levels is too small (=5), a drop in accuracy can be observed due to inadequate information capacity of the corresponding severity levels and also the lack of aggregation flexibility (like that in max pooling). Besides, the accuracy also drops when the number of levels is too large. This can be ascribed to the inability of sample-level labels to provide accurate guidance in learning clip-level severity. Optimal values of the considered metrics (set in bold) were obtained for intermediate severity level values, ranging from 10 to 20 (Table VI).

TABLE VII
SCORING RESULTS WITH DIFFERENT KEY INSTANCE PROPORTIONS

Key Instance Proportion	Acc (%)	Acceptable Acc (%)	Macro average		
			Rec (%)	Pre (%)	F1 (%)
0.1	58.38	92.43	55.24	56.93	54.29
0.2	58.15	91.98	54.60	55.22	53.34
0.3	58.89	**93.08**	55.84	55.73	54.80
0.4	59.98	91.84	**57.71**	58.33	**56.38**
0.5	**61.11**	91.85	57.33	57.95	56.31
0.6	58.72	91.75	56.96	**59.48**	55.89
0.7	59.66	91.62	56.72	58.23	55.73
0.8	58.86	91.96	55.63	57.76	54.64

TABLE VIII
PERFORMANCE COMPARISON WITH PROTOTYPE CLASSIFICATION METHOD

Models	Acc (%)	Macro average		
		Rec (%)	Pre (%)	F1 (%)
Baseline	51.54	52.47	50.25	51.54
Ours (AVG)	**58.91**	**56.19**	**57.08**	**55.02**
Baseline + DNC [61]	40.60	34.24	35.03	32.51
Ours (AVG) + DNC [61]	**51.04**	**41.41**	**44.58**	**40.77**

TABLE IX
ASSESSMENT PERFORMANCE ON THE INDEPENDENT TEST DATASET

Model	Acc (%)	Macro average		
		Rec (%)	Pre (%)	F1 (%)
Our model	58.24	57.49	55.10	55.17

2) Key Instance Proportion: In the accumulation-aware ordered multiple-instance pooling module, the key instances with the highest severity are selected based on a predefined proportion P, and are then used to determine the output of the transient stream. We varied P to explore its effect on the assessment performance. Table VII shows that $P = 0.5$ yields the highest classification accuracy. Since important clips may be discarded if the proportion is too small, while a large proportion would lead to the inclusion of inessential clips, careful selection of P is needed to ensure good model performance. Optimal values of the various considered metrics (set in bold) were obtained for intermediate values of the key instance proportion P, ranging from 0.3 to 0.6 (Table VII).

F. Results With Prototype Classification Method

To evaluate the effectiveness of the features extracted by our proposed model, we conducted experiments using a prototype classifier instead of the traditional classifier with a fully-connected layer and softmax in our model. For prototype classifiers [61], [62], a sample would be classified into the class that corresponds to the prototype closest to this sample's feature representations. Specifically, we performed an ablation study using the deep nearest centroids (DNC) method [61] for both the baseline model and our final model. The average pooling (AVG) is adopted in our final model to replace the accumulation-aware ordered multiple-instance pooling for ensuring compatibility with the DNC method. Table VIII demonstrates that our proposed method leads to consistent improvement even under the DNC classification scheme.

TABLE X
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART SKELETON-BASED ACTION RECOGNITION MODELS

Models	Acc (%)	Acceptable Acc (%)	Macro average			Param. (M)	Training Time (s)	Inference Time (s)	GFLOPs
			Rec (%)	Pre (%)	F1 (%)				
ST-GCN [17]	58.54	91.14	56.04	57.04	55.17	3.066	**818.93**	6.09	13.707
Motif-STGCN [63]	57.44	90.94	53.87	54.55	52.80	1.723	1487.99	**4.69**	**8.916**
DGNN [31]	55.03	89.95	51.74	54.96	49.85	4.048	5120.26	27.01	29.892
Ours	**61.11**	**91.85**	**57.33**	**57.95**	**56.31**	**1.605**	2487.22	13.03	9.459

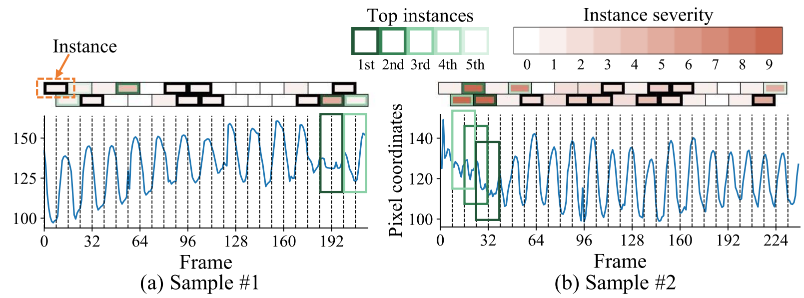

Fig. 7. Visualization of key instances. Instances are represented by blocks at the top of the figure. The color depth of a block represents the severity for the corresponding instance (the deeper the color the more severe the instance). The key instances picked by the transient stream are marked as bold frames. The top 5 key instances with the highest predicted severity are further marked with green boundaries. Remarkable regions on the curve are marked with rectangles with the same color as the corresponding key instances.

G. Independent Test Performance

We test the proposed model on the independent test dataset (described above). The results in Table IX show that the proposed model still performs well with an accuracy of 58.24% (2.87% lower than the average accuracy on the cross-validation dataset). This confirms the high stability and good clinical practicality of the proposed method.

H. Comparison With General-Purpose Skeleton-Based Action Recognition Models

We compare the performance and computation cost of our proposed model on the abovementioned cross-validation dataset against other general-purpose GCN models for skeleton-based action recognition. Table X shows that our model performs better than the models of ST-GCN [17], Motif-STGCN [63], and DGNN [31] in terms of all reported performance metrics. Furthermore, our model requires fewer parameters and giga floating-point operations per second (GFLOPs) than both ST-GCN and DGNN. Notably, the training and inference time of our model is approximately half of that of the heavily parameterized DGNN. These results demonstrate that our model achieves a favorable balance between performance and computation cost, making it more competent in assessing the PS task with reasonable computation overhead.

I. Visualization Analysis

Clip-level (i.e., instance-level) severity can be retrieved from the proposed accumulation-aware ordered multiple-instance pooling (AAOMIP) module to explore the labeling process of the transient stream. In Fig. 7, clip-level scores and the key instances picked in the transient stream are visualized to show that the correct clips are emphasized. The average values of the x coordinates of all joints of the index finger are plotted versus time to represent hand movements. Since close curve peaks or violent jittering indicate PD-induced halts, Fig. 7 shows that key instances are chosen where there are PD-induced halts (see the corresponding regions on the curve within the green frames). This demonstrates the ability of the transient stream to extract PD-related transient features.

Changes in clip-level severity scores with and without the incorporation of the proposed instance context encoding (ICE) module are also plotted to demonstrate the ability of this ICE module to suppress the effects of periodic halts. Two samples without the PD-induced halts are shown in Fig. 8, where all clips should ideally have low severity. Without the ICE module, the transient stream fails to discriminate between the natural periodic halts and the PD-induced halts, and it mistakenly assigns high severity scores to clips in a periodic pattern. Figure 8 shows that such assigned severity values are significantly decreased after the ICE module is applied, indicating that the natural periodic halts are no longer mistakenly deemed as important features related to sample-level severity. This confirms the effectiveness of the ICE module in promoting discrimination between the periodic halts and the PD-induced halts.

To sum up, equipped with the proposed AAOMIP and ICE module, our proposed model can accurately mine clips with PD-induced halts as key instances and then give reliable MDS-UPDRS scores. Finally, the positional information of these clips can be used to guide neurologists' attention and accelerate clinical decision making.

V. DISCUSSION

The MDS-UPDRS rating principles single out two prominent issues which are particularly important to the rating process: 1) which features should be considered, and 2) how to aggregate the PD-induced halts to get final scores. Specifically, 1) both persistent and transient features should be

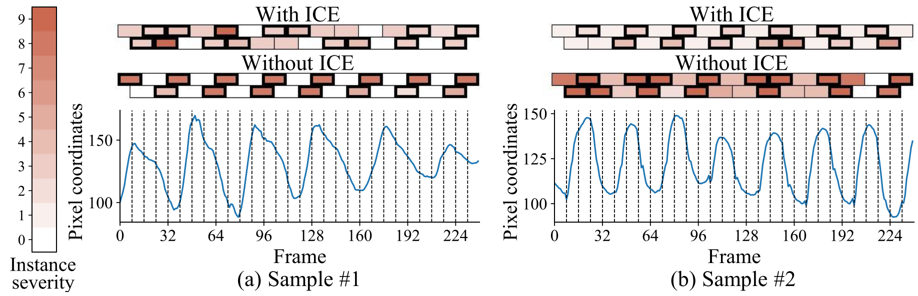

Fig. 8. Visualization of the severity levels of the clips with (the first row) and without (the second row) the instance context encoding module. The two figures in each column represent the same sample. Elements in this figure are of the same meaning as those in Fig. 7.

considered, and 2) the increase in either intrinsic severity or the number of the PD-induced halts should potentially lead to an increase in the sample-level severity. Our method is based on the MDS-UPDRS rating principles, and hence effectively simulates the clinical rating process.

Our proposed method has three main characteristics: 1) To comprehensively extract persistent and transient features, a multi-scale framework is proposed with two streams respectively focusing on persistent and transient features, where the stream for transient feature extraction is further modeled under a multiple-instance-leaning scheme in order to detect PD-induced halts. The experimental results show that the proposed framework performs better than either stream alone (Table IV). This framework can be potentially applied to other video-based tasks involving both persistent and transient features for comprehensive feature extraction. 2) To properly model the process of determining sample-level severity on the basis of PD-induced halts, we developed an accumulation-aware ordered multiple-instance pooling module in order to obtain key high-severity instances and then aggregate these instances cumulatively to estimate sample-level severity. Table V demonstrates the superiority of this module over other existing aggregation methods on all reported metrics. Additionally, the intrinsic order among instances has seldom been accounted for in multiple-instance-learning methods [64], but our experiments indicate that explicit modeling of this order can greatly boost performance. This module may inspire future multiple-instance-based works where instances' count and intrinsic order are concerned. 3) To avoid the interference of the periodic halts during the detection of the PD-induced halts, an instance context encoding module is developed to provide periodic movement information. The experimental results show that the application of this module leads to an accuracy improvement of 1.93% in comparison to the baseline model (Table IV).

Our proposed model demonstrates high clinical values. For the transient stream, key clips are chosen according to the severity levels (Fig. 7), and then sample-level severity is estimated based on the chosen key clips. The selected clips and their predicted severity values can help neurologists find out and evaluate PD-induced halts in clinical settings.

It is worth noting that our proposed model achieves an accuracy of 61.11% and an acceptable accuracy of 91.85%, with 1-point classification errors being typical. We have carefully analyzed these errors and observed that the samples with these errors often exhibit extremely similar motion characteristics to certain samples from the predicted class. This indicates that the fine-grained nature for rating the PS task leads to exceedingly subtle differences between samples from adjacent classes, thereby contributing to these errors. In our future work, we will strive to address this challenge. Moreover, we will continue to engage in the ongoing accumulation of video data from PD patients in diverse clinical states. This will contribute to the development of more robust and reliable automated assessment methods, ultimately enhancing the practical usability of the automated rating system across intricate clinical conditions.

VI. Conclusion

In this paper, we proposed a video-based assessment model for pronation-supination movements of hands. The proposed model is designed based on the MDS-UPDRS rating principles to fully exploit prior clinical knowledge. Our technical contributions are as follows: 1) a multi-scale framework is proposed to effectively extract persistent and transient features; 2) an accumulation-aware ordered multiple-instance pooling module is designed to model the multiple-instance aggregation process according to the MDS-UPDRS rating principles; 3) an instance context encoding module is proposed to provide instances with phase information of the background movements in order to avoid interference of periodic halts in the detection of PD-induced halts. The proposed model shows promising performance on both a cross-validation dataset and an independent test dataset. Thus, our proposed scheme exhibits a great potential for large-scale applications, as it only requires consumer-level cameras and is much simpler in usage than sensor-based methods.

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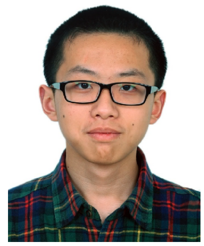

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